Supplement: Detector-Assisted Annotation Audits Capture Gaps, Count Estimation, and Review Allocation

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This supplement records the natural-reference study's target, existing estimator specializations, supporting identities, reproducibility boundaries, and complete primary-budget comparisons. The frozen specification and complete machine-readable tables accompany it. The anticipated directional penalty failed; no threshold, detector, reference policy, seed family, or method was replaced after that result. The favorable opposite-direction observation is not relabeled as success of the original harm gate.

S1 Chronology and units of evidence

The earlier Paper 1 KITTI case, synthetic comparator results, and the first OJCS interval/deletion revision were known before this study. The present study fixed three 1,000-image cohorts, two policies, two detectors, fourteen estimators, three budgets, selection/evaluation streams, and all contrasts before natural-audit performance. Its protocol is a local frozen record, not a publicly registered preregistration.

VOC train supplies development; COCO and VOC validation supply confirmation. There are four development cells and eight confirmation cells. Each cell has 10 independent synthetic-selection blocks of 20 paths. Development evaluation has 10 blocks; confirmation evaluation has 50. Every evaluation path estimates four states at three budgets with fourteen methods. Selection estimates two synthetic states at 200 reviews with fourteen methods. In total the fixed computation contains 560 blocks and 1,545,600 point estimates. These are simulation outputs, not 1.5 million independently labeled images. Shared frames and streams induce dependence that is preserved in the reported paired analysis.

All source locks and raw block hashes are retained. The point-only adapter bypasses confidence-interval inversion without changing points or complete scientific traces in qualified comparisons. Consequently this study establishes no new coverage, confidence-sequence, or interval-width result.

S2 Source identity and measurement boundary

The reference release is *Refined Annotations with Soft Labels for Popular Object Detection Datasets*, by Sarina Penquitt, Jonathan Klees, Antonia van Betteray, Parssa Jashnieh, Peter Stehr, Matthias Rottmann, and Lars Schmarje (2026). The pinned Hugging Face revision is `01d12b7bd36aae7dbb1e79ba9ee595225885e10b`; the inspected source repository revision is `6c4d09ba60a6d5041cde68a739d7526f91a02e37`. The dataset card specifies CC BY-SA 4.0. The

source article is `https://arxiv.org/abs/2609.21822`. Images retain their separate original dataset terms.

The source's object-discovery and class-review pipeline is stronger reference evidence than a single-rater marking pass. It is still incomplete and policy-dependent. Original-seeded additions are excluded to distinguish discovery provenance. The source's Appendix B.5 describes human review and class voting for accepted original-only additions, so exclusion cannot be described as proof that these records lacked human review. Class-score thresholds do not establish spatial certainty or physical completeness.

Reference box coordinates are used as released. Original VOC coordinates use $[x_{\min} - 1, y_{\min} - 1, x_{\max}, y_{\max}]$; COCO uses clipped zero-based $[x, y, x + w, y + h]$. The matching objective first maximizes cardinality subject to IoU$\geq$ 0.5, then total IoU. Source identifiers are unique in each eligible frame. Zero-area boxes are excluded and logged; a prediction-boundary validation tolerance was corrected to float 32 resolution after 22 coordinates exceeded the canvas by one float 32 unit. The maximum excess was 0.000030517578125 pixels. No prediction was clipped, discarded, or modified by that correction.

Inclusive COCO cap cases are image 122962:21 to 20;143572:115 to 20;229849:43 to 20;304404:34 to 20. Strict COCO cases are 143572:33 to 20;229849:31 to 20;304404:25 to 20. Thus the cap changes the target and is disclosed rather than treated as an incidental computational bound. There is no post-result cap sweep.

S3 Supporting identities and their limits

S3.1 Target equality

For eligible reference set R_i, let C_i be the nodes captured by original annotations under the stated matching. Normalized retention keeps C_i itself. Uniform and height controls retain a subset $S_i \subseteq R_i$ with $|S_i| = |C_i|$. Each retained node has an identity edge of IoU 1 to its source reference node, so a one-to-one matching of size $|S_i|$ exists. No matching can have more edges than retained nodes. Its maximum cardinality is therefore exactly $|S_i|$, giving the same $m_i = |R_i| - |C_i|$ and $Y_i = \min(20, m_i)$ in all states. Overlap does not invalidate this argument because distinct nodes retain their source identity.

For a common simple random image sample J, $\widehat{B}_{\mathrm{HT}} = N \sum_{i \in J} Y_i / b$ is consequently identical across states for every path. This is a pathwise negative control, not merely equality in expectation. Adaptive methods can select different images under their shared uniforms; such changes are part of their method effect.

S3.2 Fixed-predictor risk

Let $e_i = Y_i - cq_i$ for fixed c, q. Under simple random sampling without replacement, $\widehat{B}_c = \sum_i cq_i + N\bar{e}_J$ has expectation B and

$$\mathrm{MSE}(\widehat{B}_c) = N^2 \frac{1 - b/N}{b} S_e^2 = N^2 \frac{1 - b/N}{b} (S_Y^2 + c^2 S_q^2 - 2cS_{Yq}),$$

where variances and covariance use denominator $N - 1$. For $c = 1$, assistance improves on direct HT exactly when $S_q^2 < 2S_{Yq}$. At identical Y, the N/U risk difference is the common design factor times $\Delta S_q^2 - 2\Delta S_{Yq}$. These are established finite-population identities used to interpret the experiment, not new statistical theorems.

Complete-frame moments and the unrestricted oracle scalar coefficient S_{Yq}/S_q^2 require the full outcome vector. They are explanatory ceilings, not methods available before review. Plugging a fitted or changing predictor into this fixed-predictor formula does not establish its risk. Exact analytic checks in the reducer apply only to the designs for which the identity is valid.

S3.3 What deletion can do to unmatched proposals

Write $\nu(D, S)$ for maximum-matching cardinality between detector nodes D and retained reference nodes $S \subseteq R$. Deleting m nodes cannot increase ν, and can decrease it by at most m: remove edges incident to those nodes from a maximum full-reference matching. Therefore

$$0 \leq \{|D| - \nu(D, S)\} - \{|D| - \nu(D, R)\} \leq m.$$

Clipping both counts to $[0, 20]$ preserves monotonicity and cannot increase the difference. This holds for N/U/H, which are subsets of the same reference pool; it need not hold for A. A chosen maximum matching assigns capture to particular nodes, but other equally optimal assignments may exist. Per-node capture is an operational diagnostic, not a uniquely identified causal effect of deleting that node.

S4 Estimator implementation

The following definitions are fixed count implementations. They are not claims to reproduce every feature learner, tuning rule, uncertainty procedure, or dataset protocol of native PPI, Cross-PPI, StratPPI, LURE, or PPAT packages.

Direct HT and fixed difference
A uniform without-replacement sample of exactly b images gives $N\bar{Y}$. Fixed difference uses $\sum_i q_i + N\overline{(Y - q)}$. Half difference replaces q by $q/2$.

Stratified HT
Strata are unmatched proposal counts 0,1, and at least 2. Allocate $\min(N_h, 2)$ first, then the remaining budget in proportion to remaining stratum capacities, using exact largest remainders and stratum-index ties. Independently permute within strata. Estimate $\sum_h N_h \bar{Y}_h$.

Split ridge and split offset
Spend $\lfloor b/5 \rfloor$ uniform labels on fitting and add their measured total exactly once. Predict and residual-correct the remaining population with the remaining sample. Six-feature ridge uses penalty 10; nine-feature offset ridge uses penalty 100 and corrects the raw proxy. Public features are standardized; the intercept is unpenalized and predictions are clipped to[0,20].

Sequential offset
Start at q. Refit offset ridge using only previous labels after 20,40,80,120, and 160 completed draws when reached. If $M_t = N - t + 1$, A_{t-1} is the observed prefix sum and $m_{t,i}$ is predictable, define $Z_t = A_{t-1} + \sum_{i \in R_t} m_{t,i} + M_t(Y_{I_t} - m_{t,I_t})$. The estimate is $\sum_{t=1}^{b} w_t Z_t$, where $w_t = N(N-b)/(bM_t(M_t - 1))$. Conditional means equal B and weights sum to 1. A census uses direct summation.

Fitted power
Estimate sample covariance(Y, q)/sample variance(q), clip to[0,1], and use zero when the denominator is below10^{-12}. Insert this coefficient in difference correction. Because the same sample fits and scores, exact design unbiasedness is not asserted.

Stratified fitted power
Apply fitted-power correction separately within the same three strata and exact total budget as stratified HT.

Cross-fit offset and cross-fit direct
Use two fixed equal frame folds, independently sample $b/2$ images in each, and train each fold's offset predictor using the other fold's sampled outcomes. Residual-correct the population in each fold and sum. Both fold samples count once. Cross-fit direct uses zero predictions and isolates this sampling design.

Active LURE count and PPAT count
With centered public $c_i = q_i - \bar{q}$, fixed $\lambda \in \{0, 0.5, 1\}$ gives residual $r_i = Y_i - \lambda c_i$. Past-trained forecasts give the surrogate acquisition scores displayed in the main paper, with 10% uniform exploration. For remaining-set probability $p_t(i) > 0$, use

$$\widehat{B} = \frac{N}{b} \sum_{t=1}^{b} \left[1 + \frac{N-b}{N-t} \left\{ \frac{1}{(N-t+1)p_t(I_t)} - 1 \right\} \right] r_{I_t}.$$

This is the retained LURE finite-population point correction; the centered-proxy total is zero. The implementation independently checks equality to its weighted conditional-total representation. Forecast fitting, acquisition, and correction use only already acquired outcomes. Coefficients also change the acquisition probabilities, so the three active methods are not evaluated on identical samples.

The base feature vector is unmatched count, retained annotation count, detector count, and three unmatched height counts. The three added leave-one-out means occur in annotation, detector, unmatched order. With one group these are affine transformations of existing public counts, not additional independent contextual information. The frozen ridge regularization still acts on the specified expanded design.

S5 Reproducibility and verification scope

SHA 256 namespaces initialize PCG 64 permutations, fold permutations, and active uniforms. States and methods share the relevant streams. Synthetic deletion has its own hash-derived object-wise exponential races. Each raw block stores 20 paired estimate arrays and complete scientific-trace digests, plus frame, output, and frozen-source hashes. Traces reconstruct sampled identities, observed values, fits, and active probabilities. Exact digest replay is environment-dependent; portability to another linear-algebra build requires declared numerical tolerances.

The numerical environment is Python 3.12.2, NumPy 2.5.4, SciPy 1.18.1 on Windows, with two workers and single-threaded worker BLAS. The detector environment is Torch 2.8.0 with TorchVision 0.23.0, CUDA float 32 inference and pinned model-weight hashes. All predictions are retained, so reducing the saved results requires no cloud resource or image download.

Independent arithmetic checks recompute every saved summary, selector, and primary bootstrap contrast. Fixed first-path replays check full trace digests in every cell/state/method and separately reconstruct active probability vectors and point arithmetic. Count-control checks verify retained counts and unmatched-increment bounds. These checks establish computational consistency for the recorded study, not independent scientific replication or universal software correctness.

The two temporary inference pods were deleted and provider absence verified. A failed first launch and its compute time remain in the cost record. Estimated combined compute cost is

about$0.34, based on recorded lifetime and rate; the final invoice and storage charge have not been reconciled. No network volume or S 3 bucket was created. Infrastructure scripts are not required for numerical reproduction and are excluded from the review archive.

S6 Uncertainty and interpretation

Updated primary limits use two million paired bootstrap resamples of 50 blocks, preserving all 20 paths in each block, with two-sided tail allocation0.05/24. Their interpretation is approximate Monte Carlo precision conditional on these finite frames and the fixed selector. They do not quantify reference uncertainty, detector training-domain uncertainty, or variation over a population of datasets. Secondary results remain secondary; no favorable method, model, policy, or budget is promoted into a new prespecified test.

Separate selection/evaluation streams prevent selecting on the same simulation errors used for performance reporting. They do not integrate the selector's own uncertainty or make retrospective synthetic benchmarking operationally free. The protocol's 1.5 material-harm rule remains failed. The opposite-direction empirical interpretation is stated in the main paper without inventing a retrospective pass threshold.

S7 Complete actual-state budget comparisons

S7.1 COCO, policy I, Faster

Separate-stream selected method: PPAT count (0.5). Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	-3.30	235.00
Stratified HT	0.610	0.646	0.618	1.09	172.74
Fixed difference	0.223	0.192	0.207	-1.10	107.00
Split ridge	0.921	0.603	0.425	-0.93	142.03
Split offset	0.279	0.241	0.253	0.46	111.62
Sequential offset	0.225	0.196	0.198	-0.49	103.35
Fitted power	0.298	0.248	0.228	-5.21	110.91
Stratified fitted power	0.226	0.215	0.190	-1.82	99.20
Cross-fit offset	0.220	0.197	0.185	-1.22	97.57
Cross-fit direct	0.994	0.910	0.943	2.84	220.00
Active LURE count	0.222	0.188	0.158	2.28	79.85
PPAT count (0.5)	0.094	0.075	0.064	0.07	57.10
PPAT count (1)	0.158	0.135	0.133	-0.69	83.48
Half difference	0.400	0.376	0.374	-2.20	139.00

S7.2 COCO, policy I, Retina

Separate-stream selected method: PPAT count (0.5). Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	-1.99	230.00
Stratified HT	0.411	0.492	0.534	0.84	168.74
Fixed difference	0.667	0.665	0.664	-1.42	189.20
Split ridge	0.961	0.841	0.729	1.70	202.20
Split offset	0.739	0.681	0.624	1.51	186.58
Sequential offset	0.582	0.557	0.519	-0.37	168.76
Fitted power	0.707	0.677	0.666	-1.50	191.00
Stratified fitted power	0.400	0.473	0.504	-1.83	163.24
Cross-fit offset	0.562	0.539	0.447	5.34	160.16
Cross-fit direct	0.947	0.953	0.864	6.04	225.00
Active LURE count	0.744	0.553	0.355	-2.06	98.15
PPAT count (0.5)	0.277	0.214	0.131	-3.03	82.78
PPAT count (1)	0.229	0.189	0.132	-3.02	85.13
Half difference	0.806	0.806	0.805	-1.71	208.00

S7.3 COCO, policy S, Faster

Separate-stream selected method: PPAT count (0.5). Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	4.55	170.00
Stratified HT	0.647	0.753	0.753	1.92	151.64
Fixed difference	0.494	0.513	0.444	-1.55	117.00
Split ridge	0.901	0.736	0.633	-0.09	135.54
Split offset	0.498	0.493	0.423	0.90	115.49
Sequential offset	0.427	0.391	0.351	-0.50	102.06
Fitted power	0.341	0.348	0.332	-7.30	101.96
Stratified fitted power	0.259	0.290	0.272	-5.00	90.91
Cross-fit offset	0.376	0.354	0.344	2.98	102.58
Cross-fit direct	0.941	0.952	0.978	-1.00	170.00
Active LURE count	0.408	0.525	0.393	-1.66	81.02
PPAT count (0.5)	0.130	0.130	0.121	1.93	60.24
PPAT count (1)	0.288	0.285	0.262	0.29	91.02
Half difference	0.357	0.344	0.342	1.50	101.05

S7.4 COCO, policy S, Retina

Separate-stream selected method: PPAT count (0.5). Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	-3.22	175.00
Stratified HT	0.410	0.480	0.668	0.19	143.29
Fixed difference	0.690	0.640	0.661	-2.65	139.00
Split ridge	1.115	0.972	0.903	-4.03	163.05
Split offset	0.895	0.777	0.763	-3.65	146.51
Sequential offset	0.679	0.632	0.641	-2.55	135.55
Fitted power	0.754	0.697	0.686	-3.78	144.01
Stratified fitted power	0.389	0.453	0.621	-2.64	134.57
Cross-fit offset	0.604	0.582	0.608	-2.92	131.70
Cross-fit direct	0.942	0.893	1.037	-3.81	170.00
Active LURE count	0.401	0.262	0.212	-0.62	69.95
PPAT count (0.5)	0.173	0.122	0.091	-0.53	52.51
PPAT count (1)	0.213	0.171	0.143	-0.50	63.97
Half difference	0.797	0.772	0.783	-2.94	150.50

S7.5 VOC val, policy I, Faster

Separate-stream selected method: PPAT count (0.5). Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	0.63	68.00
Stratified HT	0.585	0.622	0.676	-1.08	56.42
Fixed difference	1.457	1.349	1.294	-1.84	75.00
Split ridge	1.008	0.988	0.885	-0.33	63.02
Split offset	1.410	1.118	0.944	-0.87	64.37
Sequential offset	1.110	0.868	0.755	-1.26	57.74
Fitted power	0.631	0.575	0.590	-2.10	53.33
Stratified fitted power	0.492	0.541	0.568	-3.83	51.18
Cross-fit offset	0.995	0.749	0.646	0.51	55.54
Cross-fit direct	1.114	1.033	1.018	1.84	68.00
Active LURE count	1.723	1.363	1.092	1.38	55.39
PPAT count (0.5)	0.428	0.375	0.297	0.09	35.98
PPAT count (1)	0.931	0.787	0.719	1.90	57.42
Half difference	0.585	0.572	0.570	-0.60	51.50

S7.6 VOC val, policy I, Retina

Separate-stream selected method: PPAT count (0.5). Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	1.38	67.00
Stratified HT	0.566	0.610	0.613	0.44	53.10
Fixed difference	0.748	0.749	0.750	1.03	60.00
Split ridge	1.132	1.085	1.034	0.32	68.09
Split offset	0.899	0.893	0.925	0.33	63.35
Sequential offset	0.745	0.749	0.730	1.09	56.87
Fitted power	0.734	0.725	0.729	0.94	57.20
Stratified fitted power	0.567	0.610	0.627	0.65	53.13
Cross-fit offset	0.822	0.754	0.784	0.21	60.04
Cross-fit direct	1.019	0.955	1.031	0.88	68.00
Active LURE count	2.180	1.657	1.069	-0.23	52.95
PPAT count (0.5)	0.593	0.515	0.418	-1.09	42.97
PPAT count (1)	0.560	0.471	0.415	-0.67	43.29
Half difference	0.744	0.742	0.739	1.21	58.50

S7.7 VOC val, policy S, Faster

Separate-stream selected method: Stratified fitted power. Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	-0.60	40.00
Stratified HT	0.575	0.677	0.700	-0.82	33.14
Fixed difference	5.127	5.337	5.428	-0.69	92.00
Split ridge	1.169	1.086	1.070	-1.22	41.47
Split offset	4.419	3.349	2.446	-0.61	60.24
Sequential offset	3.545	2.741	1.831	-0.64	52.88
Fitted power	0.907	0.818	0.778	-0.36	34.05
Stratified fitted power	0.682	0.746	0.742	0.08	33.62
Cross-fit offset	2.497	1.874	1.295	0.50	44.87
Cross-fit direct	0.968	0.931	0.942	-0.25	40.00
Active LURE count	2.197	1.913	1.213	-0.11	34.61
PPAT count (0.5)	0.857	0.730	0.667	-0.91	31.71
PPAT count (1)	2.521	2.534	2.127	-0.01	58.49
Half difference	1.341	1.378	1.377	-0.65	44.00

S7.8 VOC val, policy S, Retina

Separate-stream selected method: PPAT count (0.5). Values are MSE divided by direct HT at the same budget. At $b = 200$, bias and 90th-percentile absolute error are in counts.

Method	$b = 50$	$b = 100$	$b = 200$	Bias 200	Abs. error P 90
Direct HT	1.000	1.000	1.000	-0.18	40.00
Stratified HT	0.647	0.686	0.712	0.75	32.82
Fixed difference	1.460	1.441	1.409	0.11	47.00
Split ridge	1.217	1.152	1.152	-0.41	40.78
Split offset	1.627	1.486	1.279	-0.46	42.83
Sequential offset	1.301	1.157	1.048	0.05	38.67
Fitted power	0.955	0.863	0.864	0.16	35.40
Stratified fitted power	0.652	0.692	0.729	0.86	32.92
Cross-fit offset	1.177	1.127	0.961	0.53	38.18
Cross-fit direct	0.963	1.054	1.079	1.02	40.00
Active LURE count	2.841	2.234	1.456	-0.02	37.35
PPAT count (0.5)	0.634	0.605	0.457	-0.64	25.83
PPAT count (1)	0.879	0.795	0.700	0.19	32.04
Half difference	0.876	0.838	0.842	-0.03	34.20

S8 All four states at the primary budget

S8.1 COCO, policy I, Faster

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.618	0.665	0.690	0.687
Fixed difference	0.207	0.239	0.245	0.244
Split ridge	0.425	0.434	0.398	0.419
Split offset	0.253	0.279	0.252	0.268
Sequential offset	0.198	0.217	0.207	0.211
Fitted power	0.228	0.237	0.207	0.221
Stratified fitted power	0.190	0.196	0.167	0.179
Cross-fit offset	0.185	0.198	0.185	0.194
Cross-fit direct	0.943	0.943	0.943	0.943
Active LURE count	0.158	0.195	0.168	0.177
PPAT count (0.5)	0.064	0.069	0.057	0.063
PPAT count (1)	0.133	0.158	0.176	0.177
Half difference	0.374	0.341	0.291	0.309

S8.2 COCO, policy I, Retina

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.534	0.582	0.489	0.508
Fixed difference	0.664	0.619	0.389	0.474
Split ridge	0.729	0.751	0.584	0.660
Split offset	0.624	0.629	0.435	0.533
Sequential offset	0.519	0.516	0.358	0.435
Fitted power	0.666	0.624	0.393	0.479
Stratified fitted power	0.504	0.523	0.330	0.403
Cross-fit offset	0.447	0.449	0.331	0.385
Cross-fit direct	0.864	0.864	0.864	0.864
Active LURE count	0.355	0.411	0.324	0.287
PPAT count (0.5)	0.131	0.143	0.099	0.108
PPAT count (1)	0.132	0.169	0.127	0.145
Half difference	0.805	0.770	0.611	0.670

S8.3 COCO, policy S, Faster

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.753	0.822	0.833	0.823
Fixed difference	0.444	0.948	1.021	1.003
Split ridge	0.633	0.642	0.595	0.611
Split offset	0.423	0.716	0.707	0.704
Sequential offset	0.351	0.573	0.580	0.577
Fitted power	0.332	0.449	0.431	0.429
Stratified fitted power	0.272	0.373	0.363	0.363
Cross-fit offset	0.344	0.552	0.560	0.559
Cross-fit direct	0.978	0.978	0.978	0.978
Active LURE count	0.393	0.319	0.198	0.169
PPAT count (0.5)	0.121	0.171	0.143	0.152
PPAT count (1)	0.262	0.569	0.567	0.640
Half difference	0.342	0.409	0.392	0.390

S8.4 COCO, policy S, Retina

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.668	0.731	0.677	0.682
Fixed difference	0.661	0.749	0.568	0.618
Split ridge	0.903	0.898	0.752	0.803
Split offset	0.763	0.842	0.644	0.714
Sequential offset	0.641	0.668	0.511	0.566
Fitted power	0.686	0.763	0.595	0.641
Stratified fitted power	0.621	0.711	0.557	0.607
Cross-fit offset	0.608	0.669	0.528	0.587
Cross-fit direct	1.037	1.037	1.037	1.037
Active LURE count	0.212	0.196	0.146	0.280
PPAT count (0.5)	0.091	0.144	0.097	0.116
PPAT count (1)	0.143	0.253	0.237	0.244
Half difference	0.783	0.762	0.621	0.653

S8.5 VOC val, policy I, Faster

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.676	0.664	0.614	0.632
Fixed difference	1.294	1.575	1.648	1.653
Split ridge	0.885	0.884	0.858	0.846
Split offset	0.944	0.925	0.863	0.867
Sequential offset	0.755	0.786	0.769	0.752
Fitted power	0.590	0.570	0.517	0.530
Stratified fitted power	0.568	0.539	0.474	0.489
Cross-fit offset	0.646	0.656	0.597	0.594
Cross-fit direct	1.018	1.018	1.018	1.018
Active LURE count	1.092	1.015	0.673	0.826
PPAT count (0.5)	0.297	0.352	0.281	0.327
PPAT count (1)	0.719	0.802	0.866	0.961
Half difference	0.570	0.578	0.530	0.543

S8.6 VOC val, policy I, Retina

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.613	0.674	0.528	0.599
Fixed difference	0.750	0.863	0.738	0.796
Split ridge	1.034	1.052	0.970	0.956
Split offset	0.925	0.991	0.826	0.873
Sequential offset	0.730	0.785	0.674	0.713
Fitted power	0.729	0.757	0.622	0.672
Stratified fitted power	0.627	0.690	0.539	0.613
Cross-fit offset	0.784	0.789	0.640	0.676
Cross-fit direct	1.031	1.031	1.031	1.031
Active LURE count	1.069	1.514	0.723	1.056
PPAT count (0.5)	0.418	0.427	0.238	0.318
PPAT count (1)	0.415	0.508	0.364	0.416
Half difference	0.739	0.737	0.597	0.650

S8.7 VOC val, policy S, Faster

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.700	0.721	0.690	0.696
Fixed difference	5.428	7.411	7.619	7.665
Split ridge	1.070	0.994	0.945	0.950
Split offset	2.446	2.536	2.490	2.533
Sequential offset	1.831	1.949	1.902	1.928
Fitted power	0.778	0.720	0.679	0.686
Stratified fitted power	0.742	0.734	0.693	0.701
Cross-fit offset	1.295	1.147	1.067	1.084
Cross-fit direct	0.942	0.942	0.942	0.942
Active LURE count	1.213	1.428	1.294	1.376
PPAT count (0.5)	0.667	0.792	0.768	0.841
PPAT count (1)	2.127	3.258	3.320	3.296
Half difference	1.377	1.691	1.664	1.685

S8.8 VOC val, policy S, Retina

Method	A	N	U	H
Direct HT	1.000	1.000	1.000	1.000
Stratified HT	0.712	0.729	0.581	0.636
Fixed difference	1.409	1.954	1.830	1.921
Split ridge	1.152	1.084	0.930	0.977
Split offset	1.279	1.403	1.171	1.236
Sequential offset	1.048	1.095	0.917	0.973
Fitted power	0.864	0.818	0.665	0.708
Stratified fitted power	0.729	0.740	0.597	0.650
Cross-fit offset	0.961	1.032	0.831	0.912
Cross-fit direct	1.079	1.079	1.079	1.079
Active LURE count	1.456	1.804	1.138	1.526
PPAT count (0.5)	0.457	0.486	0.369	0.391
PPAT count (1)	0.700	0.985	0.915	1.044
Half difference	0.842	0.879	0.698	0.748

S9 Machine-readable completeness

The complete development and confirmation CSV tables contain every state, budget and method, with MSE, bias, median and 90th-percentile absolute error, pointwise Monte Carlo limits and analytic checks. CELL_DETAILS.json retains every selector score and declared contrast. MECHANISMS.json retains all deletion-block moments, height-bin diagnostics and ignored-region counts. These tables, rather than rounded typeset entries, are the numerical record.

S10 Secondary fixed-procedure and reference-policy analysis

This analysis follows review of the original outcomes. Its specification was frozen locally before the new sensitivity performance; it is not a new preregistered primary experiment. The fixed PPAT-half procedure was selected in all four development cells before original confirmation inspection, but this uniform reporting decision was made later. The original harm gate remains failed.

The UNCAPPED condition uses original uncapped counts with implementation bound 200. No observed outcome exceeds 115; proxy/forecast clipping and the K-dependent acquisition score change as well as the target. The IGNORE condition uses closed center-in-mask tests on original crowd/difficult boxes, removes reference and prediction boxes inside masks, removes flagged or masked originals, rematches the remainder and caps outcomes at 20. The two conditions are separate; the joint uncapped-and-excluded condition is not evaluated.

All original confirmation cells, original indexed random streams and $b = 200$ are retained. Four methods are run on 50 blocks of 20 paths in each new cell, yielding 800 blocks and 64,000 point estimates. A 48-comparison diagnostic family includes original, UNCAPPED and IGNORE fixed-method ratios to both stratified comparators in all eight cells. Updated paired block-bootstrap limits use two million resamples and two-sided quantiles $0.05/(2 \times 48)$ and $1 - 0.05/(2 \times 48)$. They are approximate conditional Monte Carlo limits. The smaller original budgets remain descriptive, and the original 12-contrast family is unchanged.

The required eight I/Faster sensitivity contrasts all favor PPAT-half with adjusted upper limits below 1. Across the entire diagnostic family, 40 support improvement, six are inconclusive and two support worsening. All entries follow; none are omitted because of direction. The existing code supplies direct/stratified exact design-MSE diagnostics and independent propensity/point reconstruction.

S10.1 Original: complete fixed-budget comparison

Frame	Policy	Detector	Target	PPAT RMSE	PPAT bias
COCO	I	Faster	600	34.973	0.075
COCO	I	Retina	600	52.173	-3.034
COCO	S	Faster	365	36.842	1.930
COCO	S	Retina	365	31.929	-0.527
VOC	I	Faster	177	22.526	0.086
VOC	I	Retina	177	26.580	-1.089
VOC	S	Faster	95	19.604	-0.906
VOC	S	Retina	95	15.887	-0.645

Frame	Policy	Detector	/Stratified HT	/Stratified power
COCO	I	Faster	0.104 [0.085, 0.127]	0.337 [0.279, 0.403]
COCO	I	Retina	0.246 [0.198, 0.306]	0.261 [0.209, 0.327]
COCO	S	Faster	0.161 [0.135, 0.193]	0.445 [0.367, 0.536]
COCO	S	Retina	0.137 [0.115, 0.163]	0.147 [0.125, 0.175]
VOC	I	Faster	0.440 [0.361, 0.533]	0.524 [0.426, 0.653]
VOC	I	Retina	0.682 [0.562, 0.837]	0.667 [0.549, 0.819]
VOC	S	Faster	0.953 [0.771, 1.175]	0.899 [0.714, 1.128]
VOC	S	Retina	0.642 [0.508, 0.822]	0.626 [0.495, 0.804]

S10.2 Uncapped: complete fixed-budget comparison

Frame	Policy	Detector	Target	PPAT RMSE	PPAT bias
COCO	I	Faster	733	49.796	-0.780
COCO	I	Retina	733	81.019	-2.234
COCO	S	Faster	394	38.577	-0.308
COCO	S	Retina	394	38.172	1.145
VOC	I	Faster	177	23.456	-0.304
VOC	I	Retina	177	26.471	-0.814
VOC	S	Faster	95	19.945	-1.056
VOC	S	Retina	95	16.627	0.436

Frame	Policy	Detector	/Stratified HT	/Stratified power
COCO	I	Faster	0.037 [0.031, 0.045]	0.066 [0.055, 0.079]
COCO	I	Retina	0.117 [0.094, 0.145]	0.116 [0.094, 0.145]
COCO	S	Faster	0.111 [0.092, 0.134]	0.346 [0.283, 0.428]
COCO	S	Retina	0.124 [0.103, 0.151]	0.129 [0.106, 0.157]
VOC	I	Faster	0.477 [0.399, 0.567]	0.568 [0.464, 0.689]
VOC	I	Retina	0.676 [0.548, 0.847]	0.661 [0.535, 0.828]
VOC	S	Faster	0.987 [0.784, 1.240]	0.930 [0.724, 1.191]
VOC	S	Retina	0.703 [0.574, 0.875]	0.686 [0.560, 0.855]

S10.3 Ignore: complete fixed-budget comparison

Frame	Policy	Detector	Target	PPAT RMSE	PPAT bias
COCO	I	Faster	315	33.712	1.054
COCO	I	Retina	315	74.971	0.749
COCO	S	Faster	191	29.201	-1.212
COCO	S	Retina	191	65.680	2.623
VOC	I	Faster	161	22.381	0.287
VOC	I	Retina	161	24.790	-1.933
VOC	S	Faster	86	17.402	0.180
VOC	S	Retina	86	16.393	1.063

Frame	Policy	Detector	/Stratified HT	/Stratified power
COCO	I	Faster	0.304 [0.245, 0.382]	0.431 [0.354, 0.531]
COCO	I	Retina	1.211 [0.900, 1.592]	1.225 [0.914, 1.606]
COCO	S	Faster	0.313 [0.255, 0.390]	0.444 [0.361, 0.545]
COCO	S	Retina	1.625 [1.156, 2.225]	1.589 [1.131, 2.171]
VOC	I	Faster	0.482 [0.385, 0.602]	0.587 [0.468, 0.747]
VOC	I	Retina	0.638 [0.511, 0.800]	0.629 [0.503, 0.790]
VOC	S	Faster	0.823 [0.677, 0.993]	0.793 [0.636, 0.982]
VOC	S	Retina	0.794 [0.657, 0.968]	0.771 [0.638, 0.940]

S11 Executable procedure and validation

The accompanying `audit_protocol.py` accepts only the public count matrix and a callback for the next reviewed image; it accepts no complete target vector. Its returned trace records measurement IDs and values, fitting IDs and selected probabilities. The interface exactly reproduces saved point and trace hashes in all eight original cells. Human reference quality remains an external requirement, not something this interface guarantees.

The new results passed independent reduction of 4,800 block-method means across original and secondary conditions, 192 independent real method-path reconstructions including 9,600 active draws, and 64 real-path invariance checks after changing unmeasured outcomes. No analytic direct/stratified MSE diagnostic exceeded the frozen six-standard-error check. These are numerical integrity checks, not new independent human replications. The full 96 method summary rows, 48 contrasts, 800 new raw blocks, reference-filtering ledgers, source locks and all failures are retained in the evidence bundle.

S12 Post-review aligned capture and allocation checks

These checks address a specified reviewer concern using existing frames, predictions, deletion controls and audit paths. They are post hoc; no new detector, reference, control distribution, performance simulation or selection search was introduced. The response specification fixes one allocation rule, all 24 reference conditions, all eight original confirmation cells and two bootstrap streams. It does not change the original failed harm criterion.

S12.1 Aligned omitted-node capture and residual moments

Let D_i^* be reference nodes captured by the fixed full-reference detector matching and $O_{i,s}$ the omitted set in state s. Capture is $\sum_i |D_i^* \cap O_{i,s}| / \sum_i |O_{i,s}|$. N is fixed; U/H use the exact 50 saved deletion sets. Per-image omitted counts, and thus the denominator, are equal across states. Capture uses uncapped nodes whereas estimation uses the declared image-level target. Public proxy changes and exact residual risks below explicitly bridge these two measurements. Ranges are descriptive across the retained controls, not inferential intervals. Node attribution can depend on equally optimal matching assignments.

Frame	Policy	Detector	N capture	U capture	H capture	N/U exact risk
COCO	I	Faster	0.483	0.658	0.612	1.019
COCO	I	Retina	0.169	0.387	0.320	1.590
COCO	S	Faster	0.596	0.780	0.759	0.943
COCO	S	Retina	0.256	0.502	0.467	1.302
VOC	I	Faster	0.593	0.759	0.718	0.965
VOC	I	Retina	0.277	0.564	0.485	1.184
VOC	S	Faster	0.663	0.815	0.791	0.966
VOC	S	Retina	0.337	0.634	0.568	1.035

All capture entries are proportions; U/H are means. Exact risk refers only to fixed difference, not PPAT. All 800,000 saved U/H feature rows and 8,000 detector/reference matchings were reconstructed. The following table retains each control's mean changed-proxy count, moment differences N minus control, exact risk ratio and descriptive PPAT risk ratio. Complete control ranges and block values are in `ALIGNED_CAPTURE_RISK.csv` and `ALIGNED_BLOCKS.csv`.

Frame	Policy	Detector	Control	Changed q	ΔS_q^2	ΔS_{Yq}	Exact N/control	PPAT N/control
COCO	I	Faster	U	82.24	-1.035	-0.528	1.019	1.222
COCO	I	Faster	H	72.20	-0.660	-0.329	0.999	1.096
COCO	I	Retina	U	97.30	-0.859	-0.990	1.590	1.442
COCO	I	Retina	H	84.04	-0.536	-0.608	1.290	1.325
COCO	S	Faster	U	51.24	-0.572	-0.204	0.943	1.193
COCO	S	Faster	H	47.96	-0.498	-0.173	0.947	1.121
COCO	S	Retina	U	66.28	-0.564	-0.533	1.302	1.492
COCO	S	Retina	H	63.06	-0.460	-0.401	1.188	1.247
VOC	I	Faster	U	28.58	-0.143	-0.058	0.965	1.250
VOC	I	Faster	H	22.92	-0.112	-0.043	0.964	1.077
VOC	I	Retina	U	44.52	-0.136	-0.096	1.184	1.795
VOC	I	Retina	H	35.06	-0.097	-0.065	1.102	1.346
VOC	S	Faster	U	17.20	-0.081	-0.022	0.966	1.031
VOC	S	Faster	H	14.38	-0.069	-0.019	0.970	0.941
VOC	S	Retina	U	26.82	-0.075	-0.043	1.035	1.318
VOC	S	Retina	H	22.54	-0.066	-0.033	1.001	1.243

S12.2 Complete allocation comparison

For each condition, the added design censuses stratum $q \geq 2$ when its size plus $\min(N_h, 2)$ in each remaining stratum is at most 200. Remaining reviews follow the original largest-remainder allocation over remaining capacities. Otherwise use the original allocation. The rule uses only public inputs but was selected after outcomes were available. Exact stratified HT variance is also its MSE; integer-moment rational arithmetic verifies every row. No oracle covariance enters the allocation. The denominator of the ratio below is exact; uncertainty comes from the PPAT numerator's 50 saved audit blocks.

Condition	Frame	Policy	Detector	Allocation MSE	PPAT MSE	Ratio [limits]
Base	COCO	I	Faster	2103.83	1223.10	0.581 [0.510, 0.658]
Base	COCO	I	Retina	7538.83	2722.01	0.361 [0.307, 0.424]
Base	COCO	S	Faster	1213.17	1357.37	1.119 [0.988, 1.250]
Base	COCO	S	Retina	2876.96	1019.46	0.354 [0.312, 0.401]
Base	VOC	I	Faster	635.95	507.42	0.798 [0.695, 0.929]
Base	VOC	I	Retina	957.44	706.49	0.738 [0.641, 0.847]
Base	VOC	S	Faster	279.50	384.33	1.375 [1.198, 1.566]
Base	VOC	S	Retina	364.87	252.38	0.692 [0.590, 0.807]
Uncapped	COCO	I	Faster	2103.83	2479.67	1.179 [1.016, 1.345]
Uncapped	COCO	I	Retina	7538.83	6564.06	0.871 [0.751, 1.008]
Uncapped	COCO	S	Faster	1213.17	1488.18	1.227 [1.068, 1.391]
Uncapped	COCO	S	Retina	2876.96	1457.10	0.506 [0.445, 0.571]
Uncapped	VOC	I	Faster	635.95	550.18	0.865 [0.747, 0.988]
Uncapped	VOC	I	Retina	957.44	700.69	0.732 [0.630, 0.845]
Uncapped	VOC	S	Faster	279.50	397.79	1.423 [1.216, 1.655]
Uncapped	VOC	S	Retina	364.87	276.46	0.758 [0.643, 0.880]
Ignore	COCO	I	Faster	1759.32	1136.50	0.646 [0.558, 0.746]
Ignore	COCO	I	Retina	4301.71	5620.59	1.307 [1.028, 1.641]
Ignore	COCO	S	Faster	1048.04	852.67	0.814 [0.702, 0.936]
Ignore	COCO	S	Retina	2768.31	4313.91	1.558 [1.130, 2.064]
Ignore	VOC	I	Faster	513.47	500.93	0.976 [0.848, 1.120]
Ignore	VOC	I	Retina	827.04	614.53	0.743 [0.647, 0.848]
Ignore	VOC	S	Faster	237.24	302.84	1.276 [1.093, 1.466]
Ignore	VOC	S	Retina	305.23	268.73	0.880 [0.770, 0.996]

S12.3 Bootstrap-tail precision and chronology

The original primary and secondary reductions used 10,000 resamples. At the 48-family tail, that places the lower endpoint near only the fifth order statistic. The current tables supersede those limits with two independently seeded streams of 1,000,000 block resamples each, pooled to 2,000,000. The 12-, 48- and new 24-contrast families retain separate Bonferroni tail allocations. They are not a single preregistered 84-test family. The new family is post hoc diagnostic evidence; selection uncertainty is not removed by multiplicity adjustment.

At the 48-family tail, the pooled endpoint is near the 1,042nd order statistic. The fixed check requires each stream's endpoints within 2% of pooled endpoints and the same above/below/crossing-one classification. All 84 contrasts meet it; seeds, stream endpoints and deviations are recorded. There were 48 million resamples across 24 cells, not 48 million new audits. The original 50 blocks per

cell remain the statistical basis. Stability addresses numerical resampling precision, not bootstrap coverage, reference uncertainty or dataset generalization.

The 48-family classification is now 40 improvements, six inconclusive and two worsening results; previously it was 39/7/2. The changed borderline comparison is IGNORE, strict VOC/Faster versus stratified HT. No practical conclusion depends on this boundary change. Historical output files remain unchanged and are explicitly superseded for these reported contrast limits.

S13 Historical interval and measurement context

A preceding controlled interval study in this research isolates uncertainty construction from point estimation. Across six primary synthetic mechanisms at 200 measured images, replacing the historical sequential interval with a qualified risk-limiting financial-audit (RLFA) construction reduces mean width by 11.3–13.8% on identical observations and forecasts. The RLFA construction is due to Shekhar, Xu, Lipton, Liang and Ramdas, UAI 2023, PMLR 216:1932–1941 (`https://proceedings.mlr.press/v216/shekhar23a.html`). Approximate paired intervals support the measured differences; conservative simultaneous finite bounds remain inconclusive. In three earlier COCO deletion populations, sequential point-estimation MSE exceeds direct HT despite narrower replacement intervals. The previous joint MSE-and-width target was not attained. Those failures remain part of the evidence.

A preceding single-rater KITTI case supplies a measurement boundary rather than a new natural-error confirmation. Its marks, class-overlap sensitivities, and unverified measurement-rate correction cannot establish physical completeness. The present multi-reviewer operational reference improves the evidence base but still does not remove all measurement uncertainty. The earlier interval proofs and aggregate results are preserved in the supporting bundle. No new interval-coverage claim is inferred from this point-estimation study, and PPAT's observed MSE advantage is not presented as an interval-efficiency advantage.

Historical interval evidence is located in `05_results/ANALYSIS_V2/primary_contrasts.csv`, `intervals.csv`, `points.csv`, and `SUMMARY.json`. Paper 1 aggregate evidence is in `01_baseline/paper1_source/anc/50_RESULTS/`, particularly `PAPER1_COMPONENTS_v02.json` and `PAPER1_SENSITIVITY_TABLE_v01.csv`. These are context for prior limitations, not independent confirmation of the present natural-reference result. Their historical statements are not promoted into the current point-estimation claims.

S14 Access to the current evidence

The supplied Supplementary Evidence ZIP has a top-level entry point and a SHA-256 manifest. It includes current manuscript source, figures, the complete pre-existing evidence package, current response analysis and resampling receipts, and a local verification program. It requires no private-repository login for reviewer use once uploaded with the submission. The private repository is a preservation mirror, not public access. The earlier reserved Zenodo DOI is not cited as a current archive. Images and model weights are excluded; source references and predictions sufficient for the numerical analysis are retained with their original terms. Human semantic verification of the reference and detector re-inference require the appropriately licensed original images.